

Problem

The source is supplying the maximum power to the load when the load resistance equals the source resistance.

- (a) True
- (b) False

Step-by-step solution

Step 1 of 2

Draw the circuit diagram.

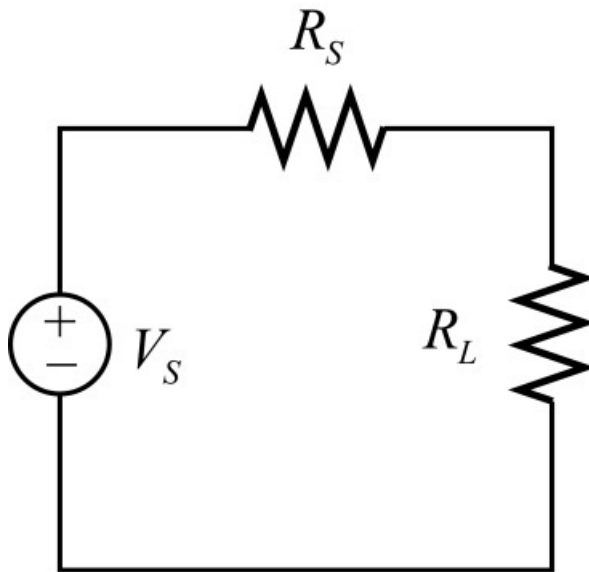


Figure 1

Step 2 of 2

The circuit in Figure 1 is Thevenin's equivalent with load resistance.

Here,

V_S is the Thevenin's voltage

R_S is the Thevenin's resistance

R_L is the Load resistance

As per maximum power transfer theorem, the Thevenin's voltage source supplies maximum power when the Thevenin's resistance is equal to the load resistance.

Thus, the statement is **True**.